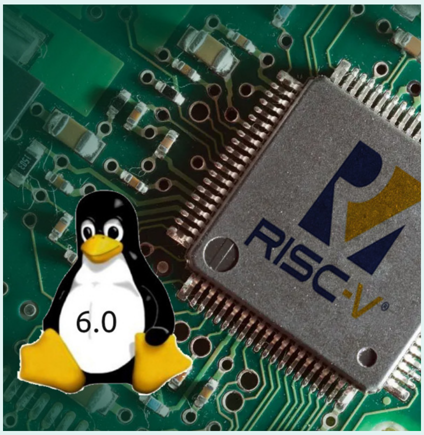
RISC-V adds kernel-mode FPU support for Linux 6.10

In the upcoming Linux 6.10 kernel cycle, the RISC-V architecture code is incorporating kernel-mode FPU (floating point unit) support. This addition is crucial for the AMDGPU kernel graphics driver, especially its DCN (Display Core Next) display code. As a result, recent AMD Radeon graphics cards should be able to function on RISC-V with display support using AMD's open source driver stack.

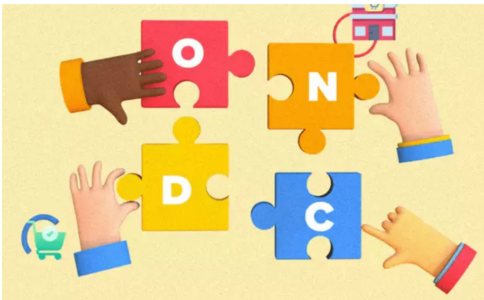
While older AMD Radeon graphics cards have been compatible with the open source AMDGPU driver on RISC-V, newer cards relying on DCN functionality have faced issues due to the lack of floating-point support. With the introduction of kernel-mode FPU for RISC-V in Linux 6.10, these compatibility issues should be resolved.



The patch enabling kernel-mode FPU for RISC-V has been queued into Andrew Morton's mm.git mm-everything branch, in preparation for the Linux 6.10 merge window next month. The patch, contributed by SiFive, provides a minimal non-preemptible implementation of floating-point code. This implementation is motivated by the amdgpu DRM driver, which requires floating-point code to support recent hardware.

Support for kernel-mode FPU is limited to the riscv64 architecture, as riscv32 requires runtime assistance (from libgcc) to convert between doubles and 64-bit integers.

Noteworthy among the brands that have joined the ONDC bandwagon are culinary giants such as Wow Momo, McDonald's, and Domino's Pizza in the F&B sector. Additionally, the network has attracted major players from the fast-moving consumer goods (FMCG) industry, including Manico, Procter & Gamble (P&G), and Hindustan Unilever Limited. Besides Namma Yatri, the Kochi Open Mobility Network and Ola have also come on board in the mobility domain.



A closer look at the order volumes and category-wise distribution reveals that the non-mobility space is diversifying. As of March this year, F&B constituted 9.55% of the total monthly orders, followed by fashion (6.74%), grocery (4.47%), and home and kitchen (3.91%). Other categories such as electronics and beauty and personal care collectively accounted for 21.89% of the orders.

The network has also seen a substantial increase in the number of sellers, from 8,500 in January 2023 to over 380,000 now, with non-mobility sectors contributing 105,000 of these. The number of network participants has also expanded from 24 to 81. Moreover, the network has broadened its scope from three domain categories to 13, enhancing its reach and inclusivity.

The impact of ONDC's expansion is also evident at the grassroots level, with the number of cities or districts generating more than 100 orders per month escalating to 622 in March. This widespread adoption is further supported by major buyer apps such as Paytm, Snapdeal, Magicpin, Pincode, Mystore, Rapidor, NoBrokerHood, Ola, and nStore, which are facilitating seamless transactions for consumers across the country.

This journey of ONDC underscores the potential of digital commerce in revolutionising the way India shops and does business, paving the way for a more inclusive and accessible digital marketplace.

Kodi 21.0, also known as Omega, released

Kodi 21.0, dubbed 'Omega', has been released, marking a significant update to the popular home theatre software. This release arrives over a year after the release of Kodi 20 'Nexus'. It brings a host of new features and improvements, including support for FFmpeg 6.0, NFSv4, reading and writing M3U8 playlists, AVIF images, and AudioEngine enhancements on Linux. Additionally, Linux users will benefit from passthrough format support such as DTS-HD and TrueHD.

Linux users will find further enhancements in Kodi 21.0. It introduces support for libdisplay-info, aiding in the parsing of EDID information to determine display capabilities. Users can now also select the audio backend using a command-line switch, providing visibility into the enabled audio backends Kodi was built with.

Other improvements include better VAAPI VP9 Profile 2 playback support, enhancements to PipeWire support, and a new command-line switch (--gl-interface=<interface>) to replace the old KODI_GL_INTERFACE environment variable usage. Raspberry Pi devices can now report CPU temperatures without relying on external scripts. Additionally, Kodi has its own implementation for ping functionality, no longer relying on Linux's ping utility.